

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, Java, JavaScript, TypeScript, C++, SQL, Bash
Systems & Testing	Spark / PySpark, Spark UI, distributed data processing, Linux profiling, performance benchmarking, integration & regression testing, pytest
Cloud & DevOps	AWS, Azure, GCP, Docker, Git, GitHub Actions, CI/CD, Cloudflare Workers (personal project)
Backend & Reliability	Node.js, Express, FastAPI, REST APIs, authentication hardening, rate limiting, audit logging, multi-tenant SaaS, VAPT remediation
Databases & AI Tools	MongoDB, PostgreSQL, Redis, SQLite, Claude Code, GitHub Copilot, Cursor

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned reliability and security for **enterprise-grade multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Built a full-stack internal annotation platform (React, Node.js, MongoDB) that automated a manual spreadsheet labeling workflow into secure role-based REST APIs, cutting manual operations by **100+ hours/month** and improving throughput by **60%**.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from JavaScript to **TypeScript** across multiple repositories, improving type safety and long-term maintainability.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Spark UI, SQL, Python

- Cut end-to-end Spark ETL runtime **~25%** on a **~1.4B-row** NYC TLC dataset by replacing shuffle joins with broadcast joins (minutes to ~0.05s), enabling Adaptive Query Execution and skew-join handling, and tuning shuffle partitions and year/month partitioning with Snappy compression.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition and column pruning, filter pushdown, and adaptive skew-join configuration; profiled stage-level timings with **Spark UI** to localize shuffle and skew bottlenecks and document a repeatable optimization workflow.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Designed a staged web-discovery pipeline, typed retrieval planning, page rerank, deterministic entity extraction with LLM fallback, and output verification, returning **cell-level provenance** for every extracted value.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation, keeping extraction predictable and review-friendly to debug.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and tool failures; conversation state persists server-side in D1 without an external database.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Engineered FastAPI backend and Electron/React frontend to run **fully offline** with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling video and audio streams locally on device.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.